What about the Wheels?

Answer No. 88

The wheels should be jacked up periodically and tested, not only for smoothness of running, but for side play as well. If in spinning a front wheel a sharp click occurs now and then and the wheel is momentarily checked, it is probable that there is a chipped or split ball in the bearing which should be removed, otherwise it may necessitate the removal of the entire bearing. A wheel in perfect adjustment should, after spinning, come to rest with the tire valve directly below the hub. Undue wear of the hub bearings is usually caused by lack of lubrication and excessive friction, due to the adjusting cone being screwed up too tight. It is a good plan to clean the bearings frequently and keep the hub well filled with grease.

How are the Wheels removed?

Answer No. 89

Front wheels: Take off hub cap, remove cotter pin and unscrew castle nut and spindle washer. The adjustable bearing cone can then be taken out and the wheel removed. Care should be taken to see that the cones and lock nuts are replaced on the same spindle from which they were removed, otherwise there is a liability of stripping the threads which are left on the left spindle and right on the opposite as you stand facing the car. Rear wheels: They should not be removed unless absolutely necessary—in which case proceed as above, then with a wheel puller remove the wheel from the tapered shaft to which it is locked with a key. In replacing rear wheels be sure that nut on axle shaft is as tight as possible and cotter pin in place. The hub caps of the rear wheels should be removed occasionally and the lock nuts which hold the hub in place tightened up. If these nuts are allowed to work loose, the resulting play on the hub key may eventually twist off the axle shaft.

How does the setting of the Front Wheels differ from that of the Rear Wheels?

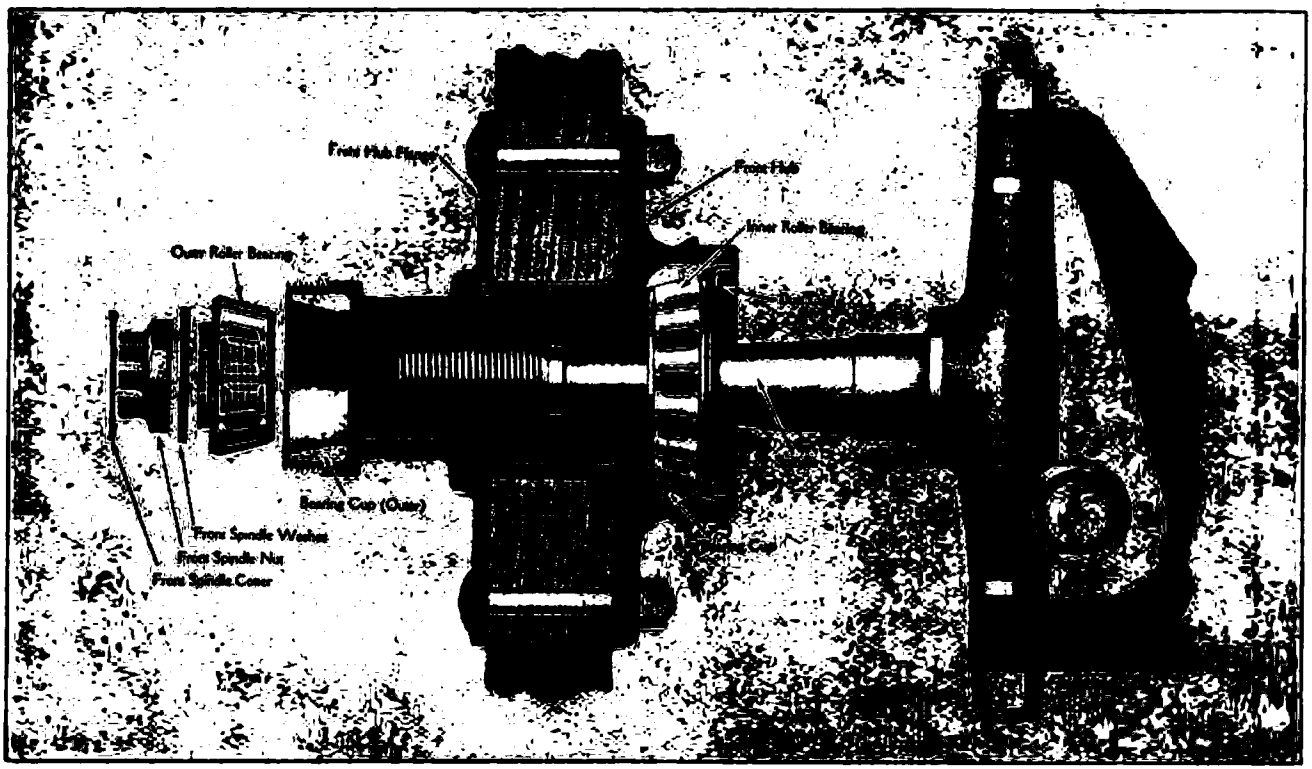
Answer No. 90

It will be observed that the front wheels are "dished;" that is, the spokes are given a slight outward flare to enable them to meet side stresses with less rigid resistance—while the spokes of the rear wheels are straight. The front wheels are also placed at an angle—that is to say, the distance between the tops of the front wheels is about three inches greater than between the bottoms. This is to give perfect steering qualities and to save wear on the tires when turning corners. The front wheels should not, however, "toe-in" at the front—at least not more than a quarter of an inch. Lines drawn along the outside of the wheels when the latter are straight in a forward position should be parallel. All wheels should always be kept in proper alignment, otherwise steering will be difficult and tire wear greatly increased. Adjustment can be made by turning the yoke at the left end of the spindle connecting rod, to draw the wheels into a parallel position.

What about installing Roller Bearing Cups?

Answer No. 91

Whenever it is necessary to install roller bearing cups, either in changing over from ball bearings to roller bearings or in replacing worn cups, this work should be done by a Dealer or Garage which has the necessary equipment for the work. In order to avoid excessive wear of the bearings it is essential that the cups be fitted absolutely true and this is practically impossible without the special equipment.



Sectional view of Hub showing Roller Bearings. (Cut No. 17)

How are the Roller Bearings installed?

Answer No. 92

First, pack the hub full of clean good quality cup grease. Take the inner cone with its rollers and pack it with grease, filling all of the space around and between the rollers. Then place the inner cone in the larger cup. Next, drive the dust ring with felt washer into the inner end of the hub so that it is flush with the end of the hub.

Place the wheel carrying the inner bearings with dust ring on the spindle. The inner cone is a one-thousandth fit, or, in other words, a slip fit on the spindle. (See Cut No. 17.) It is never necessary to force the cone onto the spindle as the cups are forced into the hub. Pack the outer or threaded cone and rollers with cup grease, filling all the space between the rollers, as was done with the inner cone. The cones are made up in right- and left-hand threads to correspond to the threads on the spindles. Care should be taken that a right-hand threaded cone is not forced onto a left-hand threaded spindle. The right-hand thread is on the left-hand side of the car, while the left-hand thread is on the right-hand side of the car. Place the cone on the spindle, running it up tight enough so that the wheel seems to bind; give the wheel a few turns to be sure that all the working parts are in perfect contact; then back off the cone ¼ to ½ a turn which will be sufficient to allow the wheel to revolve freely without end play. To determine if there is end play, grasp the spokes and shake the wheel. Do not mistake loose spindle bushings for loose bearings. Insert a cold chisel between axle and spindle to take up any play while testing the bearings. Next, put on the spindle washer and nut, drawing the nut to a firm bearing. Make sure that the cone has not been forced out of adjustment. This can be determined by giving the wheel a few turns. Insert the cotter pin which locks the nut on the spindle, fill the hub cap with grease and screw it in place on the hub.

How often should the Bearings be lubricated?

Answer No. 93

Every three or four months the wheels should be removed, the old grease taken out and the hubs and bearings thoroughly cleansed with kerosene. Then repack the hub and bearings with clean grease and readjust the bearings.

What care do the Springs need?

Answer No. 94

The springs should be lubricated frequently with oil or graphite. To do this, pry the leaves apart near the ends and insert the lubricant between them. Whenever a car is given a general overhauling, the springs should be disassembled and the leaves polished with emery cloth, afterwards packing them with graphite when reassembling. Rust can be prevented from accumulating on the springs by painting them when necessary with a quick drying black paint. You will find that these suggestions if carried out will not only improve the riding qualities of the car but prolong the life of the parts as well.

Should Spring Clips be kept tight?

Answer No. 95

Yes. If the spring clips are allowed to work loose the entire strain is put on the tie bolt which extends through the centre of the spring. This may cause the bolt to be sheared off and allow the frame and body to shift a trifle to one side. It is a good plan to frequently inspect the clips which hold the springs to the frame and see that they are kept tight.

What about the Steering Apparatus?

Answer No. 96

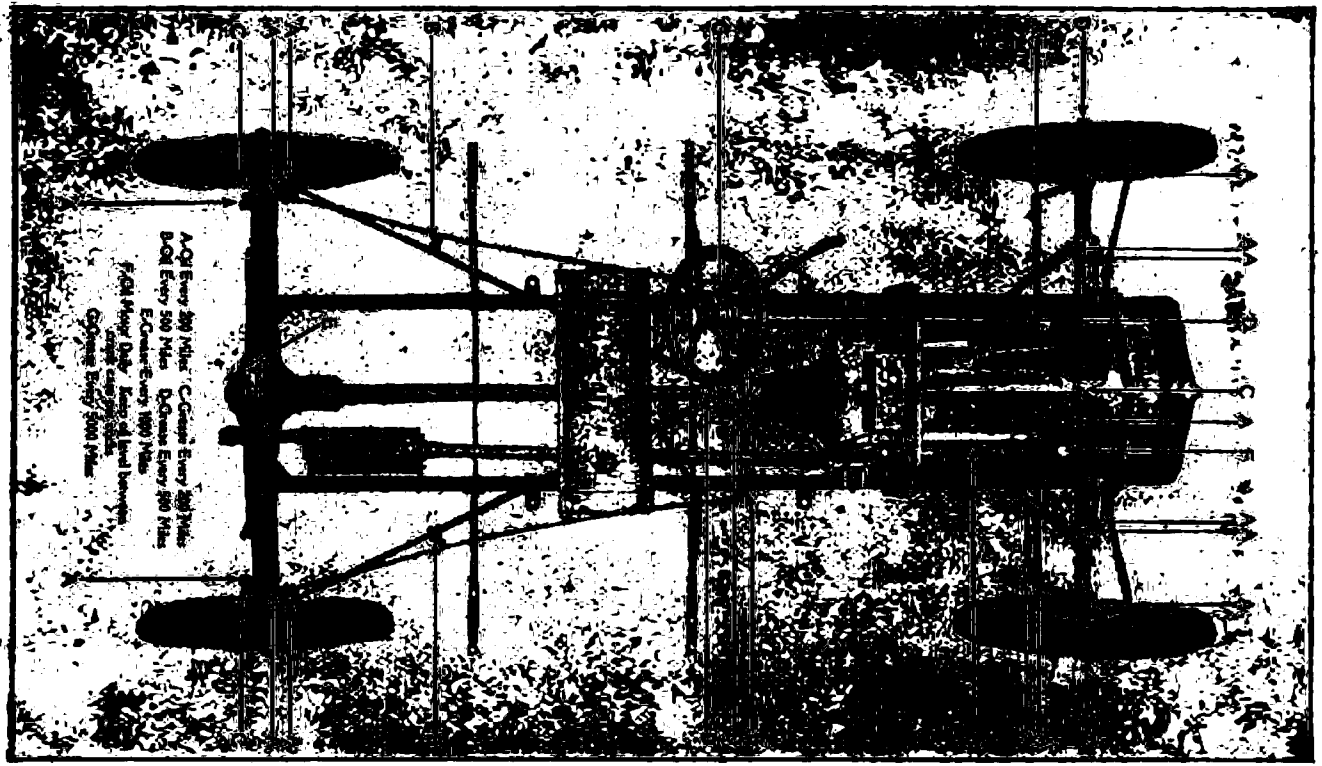
It is exceedingly simple and will need little care—except, of course, proper lubrication. The post gears which are arranged in the "sun and planet" form are located at the top of the post just below the hub of the wheel (see Cut No. 1). By loosening the set screw and unscrewing the cap—after having removed the steering wheel—they may readily be inspected and replenished with grease. To remove the steering wheel, unscrew the nut on top of the post and drive the wheel off the shaft with a block of wood and hammer.

How is Steering Gear tightened?

Answer No. 97

Should the steering gear become loose, that is, so that a slight movement of the wheel does not produce immediate results, it may be tightened in the following manner: Disconnect the two halves of the ball sockets which surround the ball arm at the lower end of the steering post and file off the surface until they fit snugly around the ball. If the ball is badly worn it is best to replace it with a new one. Also tighten the ball caps at the other end of the steering gear connecting rod in the same manner. If the bolts in the steering spindle arms appear to be loose, the brass bushings should be replaced with new ones (see Cut No. 16). Excessive play in the front axle may be detected by grasping one of the front wheels by the spokes and jerking the front axle back and forth. After the car has been in service two or three years excessive play in the steering gear may make necessary the renewal of the little pinions, as well as the brass internal gear just underneath the steering wheel spider.

It is also advisable to inspect the front spring and front spring perches occasionally to determine whether or not new bushings are necessary to overcome any excessive vibration.



Lubrication Chart. (Cut No. 18)

Don't be afraid to put a little oil into the commutator every other day—at least every 200 miles. Remember that the commutator roller revolves very rapidly, and without sufficient lubrication the parts soon become badly worn. When in this condition perfect contact between the roller and the four contact points is impossible, as a result the engine is apt to misfire when running at a good rate of speed.

Oiling the Ford Commutator. (Cut No. 19)

What about Lubricating the Differential?

Answer No. 103

Do not make the mistake of putting too much grease in the differential housing. The housing should not be more than one-third full. The differential is supplied with the required amount of lubricant when the car leaves the factory. The oil plug should be removed about every 1000 miles and more grease added if necessary. If a fluid grease is used the level should be approximately one and one-half inches below the oil hole.

Care of the Tires

How are Ford Tires removed?

Answer No. 104

First, jack up the wheel clear of the road. The valve cap should be unscrewed, the lock nut removed and the valve stem pushed into the tire until its bead is flush with the rim. This done, loosen up the head of the shoe in the clinch of the rim by working and pushing with the hands, then insert one of the tire irons or levers under the beads. The tire iron should be pushed in just enough to get a good hold on the under side of the bead, but not so far as to "pinch" the inner tube between the rim and the tool. A second iron should be inserted in the same fashion some seven or eight inches from the first, and the third tool the same distance from the second. As a clincher tire must be pried over the clinch, three levers will come in handy in a case of a "one-man job," and the knee of the driver can be used to good advantage to hold down one lever while the other two are being manipulated in working the shoe clear of the rim. After freeing a length of the bead from the clinch, the entire outer edge of the casing may be readily detached with the hands, and the damaged inner tube removed and "patched" or a spare tube inserted. Always use plenty of soapstone in replacing an inner tube.

How are Casings repaired?

Answer No. 105

Should the casing be cut so there is danger of the inner tube being blown through it, a temporary repair can be made by cementing a canvas patch on the inside of the casing. Before applying the patch the part of the casing affected should be cleaned with gasoline and when dry, rubber cement applied to both casing and patch. This will answer as an emergency repair—but the casing should be vulcanized at the first opportunity.

To prolong the life of the tire casings, any small cuts in the tread should be filled with patching cement and a specially prepared "plastic" sold by the tire companies.

How may Tire Expense be reduced?

Answer No. 106

Tire cost constitutes one of the most important items in the running expenses of an automobile. To get the most service at the least expense, the tires should be inspected frequently and all small cuts or holes properly sealed or repaired—thus preventing dirt and water working in between the rubber tread and the fabric, causing blisters or sand boils.

Tires should never be run partially deflated, as the side walls are unduly bent and the fabric is subject to stresses which cause what is known as rim cutting. The chances of getting a puncture will be greatly reduced by keeping your tires properly inflated, as a hard tire exposes much less surface to the road than a soft tire, and also deflects sharp objects that would penetrate a soft tire.

Running a tire flat, even for a short distance, is sure to be costly. Better run on the rim, very slowly and carefully, rather than on a flat tire.

Remember that fast driving and skidding shorten the life of the tires. Avoid locking the wheels with the brakes—no tire will stand the strain of being dragged over the pavement in this fashion.

Avoid running in street car tracks, in ruts, or bumping the sides of the tire against the curbing.

The wheel rims should be painted each season and kept free from rust.

When a car is idle for any appreciable length of time it should be jacked up to take the load off the tires. If the car is laid up for many months it is best to remove the tires, and wrap up the outer casings and inner tubes separately, and store them in a dark room not exposed to extreme temperatures. Remove oil or grease from the tires with gasoline. Remember that heat, light and oil are three natural enemies of rubber.

How is a puncture in the Inner Tube repaired?

Answer No. 107

After locating the puncture, carefully clean the rubber around the leak with benzine or gasoline. Then roughen the surface with sandpaper to give a hold for the cement. Apply the cement to both patch and tube, allowing it to dry for about five minutes, repeating the application twice with like intervals between for drying. When the cement is dry and sticky press the patch against the tube firmly and thoroughly to remove all air bubbles beneath it and insure proper adherence to the surface—then spread some soapstone or talc powder over the repair so as to prevent the tube sticking to the casing. Before the tube is put back into the casing plenty of talc powder should be sprinkled into the latter. A cement patch is not usually permanent and the tube should be vulcanized as soon as possible. In replacing the tire on the rim be very careful not to pinch the tube.

Points on Maintenance

What is the proper way to wash the Car?

Answer No. 108

Always use cold or lukewarm water—never hot water. If a hose is used, don't turn on the water at full force, as this drives the dirt into the varnish and injures the finish. After the surplus mud and grime have been washed off take a sponge and clean the body and running gear with a tepid solution of water and Ivory or linseed oil soap. Rinse off with cold water; then rub dry and polish the body with a chamois skin. A body or furniture polish of good quality may be used to add lustre to the car. Grease on the running gear may be removed with a gasoline-soaked sponge or rag. The nickeled parts may be polished with any good metal polish.

What care does Top need?

Answer No. 109

When putting the top down be careful in folding to see that the fabric is not pinched between the bow spacers, as they will chafe a hole through the top very quickly. Applying a good top dressing will greatly improve the appearance of an old top.

What should be done when the Car is stored?

Answer No. 110

Drain the water from the radiator, and then put in about a quart of denatured alcohol to prevent freezing of any water that may possibly remain. Remove cylinder head and clean out any carbon deposits in combustion chamber. Draw off all the gasoline. Drain the dirty oil from the crank case and cleanse the engine with kerosene as directed in Answer No. 101. Refill the crank case with fresh oil and revolve the engine enough to cover the different parts with oil. Remove the tires and store them away. Wash up the car, and if possible cover the body with a sheet of muslin to protect the finish.

What attention do the Electric Head Lights require?

Answer No. 111

Very little. When the cars leave our factory the lamps are properly focused and unless the bulb burns out there should be no occasion for removing the door, as there is nothing to get out of order. Should the door be removed for any reason care should be exercised not to touch the silver-plated reflector or the bulb with anything but a soft, clean rag, preferably flannel. To focus the lamps turn the adjusting screw in the back of lamp in either direction until the desired focus is attained.

The Ford Model T One Ton Truck

Do the Instructions relative to the Car apply to the Truck? **Answer No. 112**

The answers pertaining to the car are applicable to the truck, with the exception of Nos. 79, 80 and 81.

How are the Rear Axle and Differential disassembled? **Answer No. 113**

With the universal joint disconnected, remove the bolt in front end of radius rods and the cap screws which hold the drive shaft tube to the rear axle housing. Then remove the rear axle housing cap; also the bolts which hold the two halves of the differential housing together. With the differential exposed to view, the manner of disassembling it will be apparent. Care must be exercised to get every part back in its correct position when reassembling, being sure to use new paper liners.

How is the Worm removed? **Answer No. 114**

To remove the worm, drive out the pins which hold the coupling to the worm and drive shaft. Then remove the felt washer, roller bearing sleeve, and roller bearing by slipping them over the coupling. Drive the coupling off from the drive shaft and then force the worm from the coupling. Removing the worm nut will permit the removal of the retaining washer, thrust bearing and rear worm roller bearing. In reassembling be sure that the pin which holds the retaining washer stationary is in place.

How is the Differential Gear removed from the Shaft? **Answer No. 115**

The differential gear is fastened to the inner end of the rear axle shaft by means of splines, and is held in position by a ring which is in two halves and fits in a groove in the rear axle shaft. To remove the gear, force it down on the shaft, that is, away from the end to which it is fastened, drive out the two halves of the ring in groove in shaft with screw-driver or chisel, and force the gear off the end of the shaft.

What about Lubricating the Rear Axle? **Answer No. 116**

Extreme care must be used in lubricating the differential. An A-1 heavy fluid or semi-fluid oil, such as Mobiloil C or Whittemore's Worm Gear Protective, should be used and carried at a level with the upper oil plug. The differential is supplied with the required amount of lubricant when the truck leaves the factory and the supply should be maintained by replenishments as required. After running the truck about 500 miles, the oil should be drained off by removing the lower oil plug, and the differential filled with fresh lubricant. This operation should be repeated at approximately 1000 miles, and after that whenever necessary. The rear axle outer roller bearings are lubricated by means of dope cups. These cups should be kept filled with a good grade of grease and given a full turn every 100 miles. Before putting the truck back in service after the rear axle has been taken down, fill the differential with oil, jack up the axle and run it for five or ten minutes to insure proper lubrication of all bearings.